

Summery Post

◀ Summary Post

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Summery Post

by [Ali Alhammadi](#) - Sunday, 17 August 2025, 9:00 PM

In my initial post, I discussed Agent-Based Systems (ABS) and their increasing popularity in managing decentralized dynamic environments. I pointed out that ABS comprises autonomous agents capable of sensation, decision, and communication to attain objectives without centralized control. It makes them more flexible and dynamic than conventional systems, which tend to be rigid and depend on fixed rules (Cincotti et al., 2022). I have also briefly discussed how ABS with AI tools, such as machine learning and logic reasoning, can allow their agents to learn and change with the environment, making them best suited in industries like supply chain management, robotics, and medical.

My peers have reacted well to my post and added more points to my writing. Abdulla concurred with the adaptability and error forgiveness of ABS, highlighting this necessity in real-time-responsive industries (Zolfagharipoor & Ahmadi, 2021). His other focus was on the technology of ABS as being advanced since they could learn as time passed, courtesy of the AI technologies. Ali, in turn, enriched it by mentioning possible obstacles related to introducing ABS, specifically, the complexities of managing numerous independent agents and transparent governance (Rand & Stummer, 2021). He also argued that mixing ABS with newer technologies such as IoT or blockchain might be advisable to make them more efficient and trustworthy.

Mansour emphasized the practical applications of ABS to such areas as logistics, energy, and production, where the hybrid model may combine decentralization and supervision centralization to achieve stability (Tour et al., 2021). Another aspect that he mentioned is the significance of conflict resolution in the context of competition, stating that ABS can be enhanced using adaptive algorithms in order to guarantee that cooperation between agents is fruitful.

Finally, in conclusion, we have discussed ABS's immense potential in managing complex and dynamic environments. Despite these issues, e.g., agent coordination and governance, ABS presents a highly scalable, flexible, resilient methodology that companies can use to achieve greater efficiency and effectiveness.

References

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